Harshana Lakshara Fernando

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Summary

Full-Stack Software Engineer experienced in end-to-end application delivery, legacy integrations, and production support. Strong in **Java**, **Spring Boot**, **Kafka**, **React**, and DevOps workflows. Known for rapid learning, taking ownership of distributed systems, and choosing solutions that balance reliability, scalability, and real product needs.

Skills

Core	**Java** | **Spring Boot** | **Microservices** | Distributed Systems | System Design | **REST APIs**
Backend	Spring Cloud | Spring Security | Resilience4j | Node.js | Express.js | Gin
Messaging & APIs	**Kafka** | RabbitMQ | **gRPC** | **GraphQL** | WebSockets | Protobuf | JWT | OAuth 2.0
Data	**PostgreSQL** | MySQL | Oracle DB | Redis | Couchbase
Cloud & DevOps	AWS | **Docker** | Kubernetes | CI/CD (GitHub Actions, Jenkins)
Testing	JMeter | Gatling | Pytest | Unit Testing
Tools	Git | Gradle | Postman | Jira | Confluence

Experience

Software Engineer — Trading Systems Sep 2026 – Present
CAL *Colombo, Sri Lanka*

- Sole engineer on CAL's in-house electronic trading platform: a direct gateway to the **Colombo Stock Exchange (CSE)** plus trading screens and ops tools, replacing a vendor-built broker system.
- Carried **CSE certification** to the final stage: all application-level cases of the **188** exchange test cases re-run on CSE's test system with full exchange-message logs.
- Built an **event-sourced, CQRS** engine with a **Chronicle Queue** write-ahead log on the order path and **PostgreSQL** as an async read model, so database writes never block execution.
- Implemented **FIX 5.0 SP2** order routing and drop copy (**QuickFIX/J**) and a custom **Java NIO** decoder for the **MITCH** UDP market-data feed; built a Certification Runner that diffs every field against CSE's expected results.

R&D Software Engineer — KōreroCare Platform Mar 2026 – Aug 2026
Empathic Computing Laboratory, University of Auckland *Remote*

- Co-led backend design for the **KōreroCare** mobile-wearable sensing platform, building **Go/Gin**-based services for time-series ingestion and cross-device synchronization.
- Designed **PostgreSQL** schemas for participants, studies, devices, consent, reminders, and wearable sensor streams.
- Integrated **Keycloak OAuth2/OIDC** with **Traefik** gateway routing and Docker Compose deployment for secure internal service access.

Co-founder & Lead Engineer (Freelance) Jul 2025 – Aug 2026
ScaleX — SaaS Development Lab *Remote / Sri Lanka*

- Co-founded ScaleX and owned the end-to-end lifecycle of multiple SaaS MVPs, from architecture through cloud-native deployment.
- Built **AgriNova** (**Next.js**, **Prisma**, **Supabase**/**PostgreSQL**), an inventory and commission platform across 5 branches and 329+ SKUs, tracking 194K+ units and 860+ invoices (Rs. 64.4M+).
- Cut repeat dashboard data-fetch latency from ∼384ms to <1ms with **React Query** caching and pagination.

Software Engineer Intern — ISA Aviation Technology Dec 2024 – Jun 2025
Information Systems Associates (Pvt) Ltd *Colombo, Sri Lanka*

- Owned design and delivery of a distributed **Spring Boot**/**Kafka**-backed PNR ID generation service, **live in production**, achieving **25+ PNRs/sec** at **<100ms latency**.
- Designed a **pool-based PNR allocation model** replacing legacy Oracle-sequence generation, with **concurrency-safe** Kafka-backed allocation and Java **ReentrantLock** critical sections.
- Led the rollout to AeroOrder using database-driven **feature toggles** and a **strangler-pattern** migration, bridging the **Java 8** AeroMart monolith via **gRPC** clients and Kafka consumers.
- Resolved production **Kafka consumer lag** and offset-reset issues with graceful shutdown hooks, retry handling, validation, and a Producer–Consumer refactor.

- Built load testing and monitoring with **JMeter**, **Prometheus**, and **Grafana**; maintained **>90% unit test coverage**.

Open Source Contributions — See all

- **kgateway #14851** (CNCF) *Merged, Oct 2026*
  Added the **kgateway_build_info** Prometheus metric (version, revision, Go version, OS/arch) so operators can see which build each pod runs; shipped in **kgateway v2.5.0-beta.2**.
- **kubernetes-mcp-server #1482** *Merged, Oct 2026*
  Fixed flaky HTTP server startup in the **Go** test suite with a 5 s readiness poll and cleanup before failing.
- **kubernetes-mcp-server #1549** *Merged, Oct 2026*
  Test that every tool, prompt and resource across all 12 toolsets declares valid **Kubernetes RBAC** metadata; approved on the first review.
- **Kubeflow Spark Operator #3249** *In review (LGTM)*
  20 **Ginkgo** specs covering every executor pod-phase transition: **100%** of `sparkpod_metrics.go`, plus a log message fix.
- **KEDA #8317** (CNCF) *In review*
  New **onlyReadyPods** option for the Kubernetes workload scaler (count only Running and Ready pods); design agreed with a maintainer, docs in keda-docs #1919.
- **kubernetes-mcp-server #1568** *Root-caused, Oct 2026*
  Showed a reported "flaky" test was a real merge-order bug; the author confirmed and applied the suggested fix.

Selected Projects

AgriNova Inventory & Commission Platform (Freelance, ScaleX) — Next.js, Prisma, PostgreSQL, React Query

- Delivered a multi-branch inventory and commission platform for an agricultural business: 5 branches, 329+ SKUs, 194K+ units tracked and 860+ invoices processed, with transaction-safe stock movements.
- Replaced application-layer aggregation with **PostgreSQL** aggregations and added **React Query** caching, cutting repeat dashboard latency from ∼384ms to under 1ms; end-to-end tested with Playwright.

Employee Management System — Java, Spring Boot, Spring Cloud, Kubernetes

- Built cloud-native **Spring Boot** microservices using Eureka, Config Server, API Gateway, RabbitMQ-based configuration refresh, and Resilience4j fault tolerance.
- Added CI/CD, monitoring, Docker deployment, and Kubernetes-based architecture to improve scalability.

GoTogether Travel Platform — Next.js, Go, Spring Boot, Keycloak, PostgreSQL

- Built a full-stack travel platform with Next.js interfaces, Spring Boot social services, and **Go/gRPC** itinerary APIs.
- Implemented **OAuth2/OIDC** authentication with Keycloak and deployed services using Docker, NGINX, Certbot, and Oracle Cloud.

Event-Driven Food Delivery Platform — Java, Spring Boot, React, Kafka, Redis

- Built a full-stack food delivery platform with a React frontend, Spring Boot services, JWT authentication, and Redis caching.
- Implemented real-time order synchronization and notifications using **Kafka**, WebSockets, and service-to-service APIs.

Serverless Auction Platform — Node.js, AWS Lambda, DynamoDB, EventBridge

- Designed event-driven auction, authentication, and notification services using AWS Lambda, API Gateway, EventBridge, SQS/SNS, and DynamoDB GSIs.
- Secured APIs using a custom JWT Lambda Authorizer and improved reliability through validation and centralized error handling.

ZK-Rollup L2 Benchmarking Framework — Rust, Solidity, Python, Docker

- Built an end-to-end benchmarking pipeline with a Rust submitter, Solidity L1 contracts, and Python metrics and data-analysis tools.
- Increased submitter test coverage from **41% to 88%** and measured approximately **70% lower** data-availability cost using blob-based mode.

Maison Vella — E-Commerce Admin Panel — Next.js, tRPC, Prisma, Sanity

- Built a responsive e-commerce administration dashboard using Next.js 14, Server Actions, ShadCN UI, Tailwind CSS, and Clerk authentication.
- Implemented type-safe data access using **tRPC**, Prisma, Supabase/PostgreSQL, Sanity CMS, and Cloudflare R2 media storage.

Education

B.Sc. (Hons) in Engineering – Computer Science and Engineering *2022 – 2026*
University of Moratuwa, Sri Lanka

G.C.E. Advanced Level (Physical Science Stream) *2019*
St. John's College, Panadura, Sri Lanka
Results: 3 A's (Combined Mathematics, Physics, Chemistry) **District Rank:** 17

Training & Certifications

CKA Training — KodeKloud, 2025 • AWS SAA-C03 Training — Udemy, 2025 • Apache Kafka Series — Udemy, 2025 • 100 Days of DevOps — KodeKloud, 2025 • Advanced OAuth Security — Udemy, 2024

Articles

See all articles

- **The Singleton Design Pattern in Java: From Beginner to Production-Ready** — Software Engineering / Java
- **The Ultimate Guide to Software Optimization: Bentley's Rules Modernized** — Performance
- **The Architecture of Trust: The Ultimate Guide to API Security** — API Security
- **How Kafka Cheated the Two Generals' Problem to Achieve Exactly-Once Semantics** — Distributed Systems

Referees

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